

MinKyu Kang

Data Scientist | Data Analyst | ML / Data-driven Problem Solver

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Portfolio | GitHub

Professional Summary

Data science and software student with hands-on experience in NLP, recommender systems, data quality management, research data engineering, public data analytics, and digital twin-based manufacturing simulation. Strong at defining ambiguous problems, structuring messy data into usable pipelines, and validating results through experiments and service-oriented prototyping. Interested in building reliable data systems and intelligent models that connect analysis to real-world decision making.

Education

Sungkyunkwan University (SKKU)

Seoul, South Korea

B.S. in Data Science & Software-related interdisciplinary training

Relevant focus: Machine Learning, NLP, Recommender Systems, Data Analysis, Research Data Management

Experience

Research Assistant, Institute of Information Management, Sungkyunkwan University Sep 2024 – Dec 2025

- Participated in the KRM humanities & social science research database construction project and handled metadata inspection, cleansing, and standardization for books, papers, and survey data.
- Managed consistency of core metadata fields under an FRBR-based structure, including title, abstract, metadata-writer abstract, table of contents, and subject terms.
- Defined rule-based quality criteria for missing required fields, topic inconsistency, and duplicate expressions to improve reproducibility and data reliability.
- Systematized mismatch cases across project-raw data-intermediate outputs and documented evidence sources and page-based justification rules.
- Supported metadata-based automatic topic classification validation and assessed applicability of the model to research data management DB operations.

Research & Academic Activities

Undergraduate Researcher, DXLab, Sungkyunkwan University

Sep 2024 – Dec 2024

- Participated in multimodal harmful-content detection research combining image and text.
- Built and refined datasets and tuned multimodal models including LLaVA / KoLLaVA-based approaches.
- Evaluated model performance and feasibility for service application.

Co-author Presentation, 31st Summer Conference of the Korea Society for Information Management Jul 2024 – Aug 2024

- Presented research on metadata-integrated text design and automatic topic classification for KRM research data.
- Trained and evaluated RoBERTa-based models to verify operational applicability to research-data DB systems.

K-Software Empowerment Bootcamp (KSEB), 3rd Cohort

Mar 2024 – Aug 2024

- Completed practice-oriented AI, data, and software training.
- Developed a Korean body-data-based clothing size recommendation service, covering data preprocessing, ML modeling, and backend implementation.

AI Convergence Summer Boot Camp

Jul 2023 – Aug 2023

- Completed intensive training in AI application development, Git/GitHub collaboration, and open-source project workflows.

Selected Projects

CEVSim: Cosmetic Emulsification Vessel Scale-Up Simulator

Mar 2026 – Jun 2026

Smart Factory Capstone Design Project

- Developed a web-based digital twin simulator for cosmetic emulsification processes using Three.js and Population Balance Modeling (PBM).
- Implemented a 65-node Fixed-Pivot PBM engine to predict droplet size distribution (DSD), viscosity, hardness, and emulsion uniformity throughout the emulsification process.
- Built a real-time 3D simulation environment reproducing homogenizer, paddle, and scraper impeller dynamics with particle-based flow visualization.
- Designed a multi-metric quality evaluation framework integrating D_{32} , H-Index, viscosity, and hardness, enabling automated PASS / REVIEW / FAIL classification.
- Developed scale-up prediction logic for 100L, 300L, and 500L manufacturing vessels based on constant energy dissipation rate (ϵ) principles.
- Validated simulation outputs against real manufacturing batches and demonstrated applicability to smart manufacturing and process optimization.
- Submitted an Industry Track paper to ICSOC 2026 titled "*Scale-Up Simulation of Cosmetic Emulsion Manufacturing Vessels (100–500 L) Using a Population Balance Model: Integrated Multi-Metric Analysis of Droplet Size Distribution, Uniformity, Viscosity, and Hardness*".
- Filed a patent application entitled "*Method and System for Predicting and Evaluating Scale-Up Emulsification Quality of Cosmetic Emulsion Manufacturing Vessels Using Population Balance Models and Multi-Quality Indicators*" (Reference No. R-2026-0578-KR-1, Filed Jun. 2026).
- Registered the industrial software *CEVSim (Cosmetic Emulsification Vessel Scale-Up Simulator)* for software copyright protection.

LLM-based Personalized Policy Recommendation System

Sep 2025 – Dec 2025

Software major project

- Designed a hybrid retrieval architecture combining SQL hard filtering, vector search, and LLM re-ranking.
- Structured policy data into JSON context to enable grounded recommendations and reduce hallucination risk.
- Optimized the end-to-end RAG pipeline to improve retrieval quality and recommendation precision.
- Built a recommendation flow that links semantic search and user conditions to practical policy application behavior.

GNN-based Personalized Food Recommendation System

Sep 2025 – Dec 2025

Software major project

- Built an integrated User–Item–Context dataset by combining crawling and survey data.
- Designed a graph-based recommender with embedding propagation and hybrid ranking using CB, CF, graph score, and meta features.
- Optimized recommendation scoring with trade-offs among distance, price, and preference.
- Improved NDCG from 44.82% to 81.55% and achieved precision up to 80%.

Real-time Mental Health Detection via Chatbot

Sep 2024 – Dec 2024

Capstone project, Data Science

- Crawled and processed online counseling text data from Naver Knowledge iN.
- Designed an NLP preprocessing pipeline using TF-IDF and 2–3 gram filtering to extract top informative sentences.
- Built a labeled dataset covering 16 disorders and 48 symptoms, with an additional *Uncertain* label

for ambiguous cases.

- Applied uncertainty-aware modeling and implemented a chatbot-style prototype for real-time mental-health state classification.

Metadata-based Automatic Topic Classification for Research Data 2024

- Reorganized 45 subject classes into 35 classes to address class imbalance in KRM monograph data.
- Compared SVM, Random Forest, XGBoost, and RoBERTa for metadata-based classification.
- Achieved best performance with RoBERTa (Accuracy 0.792, F1-score 0.775).
- Verified the practical value of metadata integration for improving consistency in research-data management.

“Coolness” Night Tourism Map for Seoul Jul 2025 – Oct 2025 *Korea Tourism Data Contest project*

- Defined nine qualitative tourism-experience factors and designed a quantifiable coolness index.
- Integrated visitor and spending data to compute normalized weighted scores.
- Conducted clustering analysis by tourist-site type and implemented map-based filtering for user exploration.
- Verified service effectiveness with improved perception of coolness, local experience, convenience, and safety.

Meal Desert Solution for Elderly Food Access 2025

- Addressed structural food-access issues for elderly populations with limited mobility and information access.
- Used public and spatial data to design adjusted demand indicators for meal support.
- Applied optimization logic to propose efficient support-point placement and improve access equity.

Unsmile Classifier: Korean Hate Comment Classification 2023

- Constructed a Korean comment dataset and redefined labels into nine hate / clean classes.
- Developed a KcELECTRA-based hate-speech classifier with about 83% performance.
- Presented the outcome at the DScover academic conference.

Turnover Signal Prediction using Labor Statistics and Job-platform Reviews 2025

- Participated in a KOSSDA student competition project combining labor panel data and job-platform reviews.
- Analyzed how salary, work hours, and employment type relate to turnover signals.

Korean Body-data-based Clothing Size Recommendation Service 2024

- Built a recommendation service using Korean body-shape data for apparel sizing.
- Contributed to preprocessing, machine learning model design, and backend server implementation.

Awards

- Encouragement Award, Korea Tourism Data Contest (2025)
- Chairman’s Award, National Happiness Service Discovery / Startup Competition, Korea National Council on Social Welfare (2025)
- Excellence Award, ICT Solution Division, Creative Convergence Talent Sharing & Collaboration Festival (2024)
- Silver Award, DScover Academic Conference, Sungkyunkwan University (2023)
- Academic Excellence Scholarship, Sungkyunkwan University (2023-1)
- Golden Glove, Seoul University Football Club Federation SUFA B League (2023)

Leadership & Extracurricular Activities

President, DScover Data Analysis Society Mar 2023 – Aug 2024

- Led curriculum planning, study operations, project-based learning structure, and organization man-

agement.

- Coordinated sessions, invited talks, and collaborative projects to strengthen members' practical analytics capabilities.

Treasurer / Goalkeeper, SKKU Central Football Club

- Managed budgeting, membership fees, and administrative coordination while supporting training and match scheduling.
- Contributed as the starting goalkeeper and received the SUFA B League Golden Glove award.

Skills

Programming / Data: Python, SQL, R, JavaScript, HTML, Git, GitHub

ML / AI: Machine Learning, NLP, Recommender Systems, Classification, Uncertainty-aware Modeling, Multimodal AI

Simulation / Digital Twin: Digital Twin, Process Simulation, Three.js, Particle-based Simulation, PBM, Quality Prediction

Data Work: Data preprocessing, crawling, metadata standardization, data quality management, visualization, experimental evaluation

Product / Service: Service-flow design, prototyping, dashboard design, competition-oriented business planning

Certifications

- SQLD
- Computer Specialist in Spreadsheet & Database, Level 2